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Open Geospatial Consortium

# OGC EMISSION EVENT MODELING LANGUAGE (EMISSIONML) PART I: DATA MODEL

## DRAFT STANDARD

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# I ABSTRACT

This OGC Standard defines EmissionML, a conceptual data model for representing emission events. EmissionML provides a domain-specific ontology for describing the release of substances into the atmosphere, including the source from which an emission originates, the mechanism by which it occurs, the quantity of substance released, and the temporal bounds of the event.

EmissionML distinguishes itself through three key design principles. First, it incorporates an evidence chain linking emission determinations to supporting Observations, enabling traceability from reported values to underlying measurements. Second, it provides explicit uncertainty provenance by attaching data quality metadata to both emission quantities and temporal bounds, ensuring that the confidence and precision of each determination are recorded alongside the values themselves. Third, it offers clear spatial and temporal context — spatial through Feature (as the ultimate feature of interest) and Source (as the emission origin), temporal through start time, end time, and the phenomenon time of supporting Observations. By adopting the OMS 3.0 Observation model, EmissionML provides a common framework for reconciling heterogeneous sensor observations — from remote sensing satellites to ground-based continuous monitoring stations — within a single, unified evidence chain.

The model builds on foundational OGC and ISO standards. Source inherits from Feature (ISO 19109), DeterminationProcedure inherits from Procedure (ISO/OGC 19156:2023), and Observations conform to the OMS 3.0 Observation model (ISO/OGC 19156:2023). The standard defines twelve requirements specifying the structure and constraints of the core data model.

By making emission semantics, evidence chains, and uncertainty provenance explicit and machine-verifiable, EmissionML enables credibility of reported emission values, traceability from determinations to their supporting observations and procedures, and reproducibility of emission assessments across organizations and jurisdictions.

The intended audience includes environmental regulators, greenhouse gas reporting frameworks, emissions monitoring system developers, and geospatial data infrastructure implementers who require a standardized, interoperable representation of emission event data.

# II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, EmissionML, emission event, atmospheric emissions, greenhouse gas, air quality, emission source, emission quantity, observation, data quality, uncertainty provenance, evidence chain, interoperability, net-zero, methane emissions, sensor web

# PREFACE

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# IV SECURITY CONSIDERATIONS

No security considerations have been made for this Standard.

# V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- SensorUp Inc.
- Natural Resources Canada
- University of Calgary
- Fraunhofer
- DataCove e.U.
- GTI Energy
- Mbrace Energy
- Alberta Energy Regulator

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1

# SCOPE

# 1 SCOPE

This document defines a conceptual data model for emission events, for features from which emissions originate and mechanisms by which they occur, and for observations and procedures that provide evidence for emission determinations. These provide models for the exchange of information describing emission events and their associated quantities, temporal bounds, sources, and mechanisms, both within and between different regulatory, scientific, and technical communities.

Emission events commonly involve observations of an ultimate feature-of-interest at varying spatial scales. This document defines a common set of classes for describing emission sources, mechanisms, quantities, temporal bounds, determination procedures, and supporting observations. The schema includes relationships between emission events and their evidence chains, linking reported values to underlying observations and procedures, and attaches data quality metadata to emission quantities and temporal bounds.

This document concerns only externally visible interfaces and places no restriction on the underlying implementations other than what is needed to satisfy the interface specifications in the actual situation.

2

# CONFORMANCE

# 2 CONFORMANCE

This Standard defines a conceptual data model for emission events.

Requirements for one standardization target type are considered:

- Data Model

Conformance with this Standard shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site.

In order to conform to this OGC® Standard, a software implementation shall choose to implement:

- The Emission Event Conceptual Data Model Core conformance class specified in Annex A (normative).

All requirements-classes and conformance-classes described in this document are owned by the Standard(s) identified.

3

# NORMATIVE REFERENCES

# 3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.


ISO: ISO 19101-1:2014, *Geographic information — Reference model — Part 1: Fundamentals*. International Organization for Standardization, Geneva (2014). https://www.iso.org/standard/59164.html.

ISO: ISO 19107:2019, *Geographic information — Spatial schema*. International Organization for Standardization, Geneva (2019). https://www.iso.org/standard/66175.html.

ISO: ISO 19108:2002, *Geographic information — Temporal schema*. International Organization for Standardization, Geneva (2002). https://www.iso.org/standard/26013.html.

ISO: ISO 19109:2015, *Geographic information — Rules for application schema*. International Organization for Standardization, Geneva (2015). https://www.iso.org/standard/59193.html.

ISO: ISO 19157:2013, *Geographic information — Data quality*. International Organization for Standardization, Geneva (2013). https://www.iso.org/standard/32575.html.

ISO: ISO 19156:2023, *Geographic information — Observations, measurements and samples*. International Organization for Standardization, Geneva (2023). https://www.iso.org/standard/82463.html.

4

# TERMS AND DEFINITIONS

# 4 TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word "shall" (not "must") is the verb form used to indicate a requirement to be strictly followed to conform to this Standard and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

| | |
|---|---|
| Determination Procedure | A Determination Procedure describes the steps performed to determine the quantity or temporal bounds of an Emission Event. |
| Emission Event | An Emission Event represents the release of a substance into the atmosphere over a non-zero duration. It is characterized by its start time and end time, the Source Feature from which the substance originated, and the identity and quantity of the emitted substance. |
| Emission Quantity | An Emission Quantity is the physical quantity representing the total mass of substance released to the atmosphere during a defined Emission Event or over a specified time interval. |
| Emission Intent | Emission Intent is a code list describing whether an emission is intentional, unintentional, or unknown. |
| Feature | Abstraction of real world phenomena (Source: ISO 19101-1:2014) |
| Measurement | Set of operations having the object of determining the value of a quantity (Source: ISO/OGC 19156:2023) |
| Mechanism | A Mechanism is a classification of the specific physical, chemical, or operational process that constitutes the direct cause or pathway for an emission. It is defined independently of the Source—including its identity, structure, or purpose—allowing Mechanisms to be consistently reused across different types of Source. |
| Observation | Act carried out by an observer to determine the value of an observable property of an object (feature-of-interest) by using a procedure, with the value provided as the result (Source: ISO/OGC 19156:2023) |
| Procedure | Description of steps performed (Source: ISO/OGC 19156:2023) |
| Source | A Source represents the feature from which an emission originates. A Source is an ISO 191xx feature whose geometry provides the geospatial reference for one or more Emission Events. A Source is |

| | |
|---|---|
| | characterised by its own identity, geometry, intended function, and lifecycle, and is explicitly agnostic to the Mechanism by which any associated Emission Event occurs. |
| Source Type | A Source Type classifies a Source by its category. |
| Temporal Bound | A Temporal Bound is a value object representing a temporal determination (e.g., start time or end time) that MAY have associated data quality metadata. |
| Unit of Measure | Reference quantity chosen from a unit equivalence group (Source: ISO/OGC 19156:2023) |

5

# KEYWORDS

# 5 KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, EmissionML, emission event, atmospheric emissions, greenhouse gas, air quality, emission source, emission quantity, observation, data quality, uncertainty provenance, evidence chain, interoperability, net-zero, methane emissions, sensor web

6

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- DataCove e.U., Austria
- GTI Energy, USA
- Mbrace Energy, Canada
- Alberta Energy Regulator, Canada

7

# CONVENTIONS

# 7 CONVENTIONS

## 7.1. Overview

This section provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of XML schema, or special notes regarding how to read the document.

## 7.2. Identifiers

The normative provisions in this Standard are denoted by the URI

http://www.opengis.net/spec/emissionml/1.0

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

8

# BACKGROUND AND MOTIVATION

# 8 BACKGROUND AND MOTIVATION

## 8.1. Overview

Emissions measurement, monitoring, reporting, and verification (MMRV) now spans satellite remote sensing, aerial surveys, continuous in-situ monitoring, and handheld inspection instruments. Each modality generates data at different spatial and temporal scales, yet regulators, operators, and verifiers must reconcile these heterogeneous observations into coherent, auditable emission assessments. The absence of a common ontology and data model for emission events is the primary structural barrier to achieving this reconciliation at scale.

EmissionML addresses this gap. It defines a standards-grounded conceptual model that provides a consistent semantic foundation for describing emission sources, events, quantities, and supporting observations — independent of the sensing technology, reporting framework, or enterprise system that produces or consumes the data.

## 8.2. Structural Gaps in Current Approaches

The emissions data ecosystem lacks a shared ontology. Each technology vendor, facility operator, and reporting program defines its own vocabulary, formats, and assumptions. This fragmentation produces systemic inefficiencies across the entire data lifecycle:

- **Costly, manual, and siloed workflows.** Operators must build bespoke integrations to reconcile proprietary vendor outputs with internal enterprise systems, duplicating effort for each new sensor or data source.
- **Inability to link heterogeneous measurements.** A satellite-detected concentration anomaly and a ground-level inspection of the same source cannot be programmatically correlated when source features, event semantics, and observation metadata are defined inconsistently.
- **Incompatible reporting requirements.** The same emission data must be restated in multiple incompatible templates to satisfy different regulatory and voluntary frameworks (e.g., EPA, EU Methane Regulation, OGMP 2.0, MiQ), increasing administrative burden and undermining cross-framework comparability.
- **Lack of transparency and auditability.** Without a standard evidence chain linking reported quantities to their supporting observations and determination procedures, stakeholders cannot independently verify the credibility of emission assessments.

These are not incidental inefficiencies. They are consequences of a missing data foundation: no common model exists for the emission event itself — the central entity that connects a source, a mechanism, a quantity, a time interval, and the observations that support the determination.

## 8.3. How EmissionML Addresses These Gaps

EmissionML operationalizes open standards to provide a sensor-agnostic, interoperable data model for emission events. It establishes:

- **A shared ontology for emission events and their constituent parts** — Source, Mechanism, Emission Quantity, Temporal Bound, Emission Intent, and Determination Procedure — each with well-defined semantics and relationships.
- **An explicit evidence chain** linking every emission determination to its supporting Observations (ISO/OGC 19156:2023, OMS 3.0) and procedures, enabling end-to-end traceability from reported values to underlying measurements.
- **Embedded uncertainty provenance** through data quality metadata attached to emission quantities and temporal bounds, so that the confidence and precision of each determination are recorded alongside the values themselves.
- **A "write once, use many times" intermediary model** that decouples data production from reporting consumption, enabling a single standards-conformant emission event record to be transformed into outputs for multiple regulatory and voluntary frameworks.

Figure 1 illustrates this architecture. Heterogeneous inputs — emission inventories, sensor observations, operational data, geospatial data, and field inspection and repair data — are normalized through EmissionML into a single standards-conformant representation. From that representation, outputs are generated for multiple downstream consumers without requiring source-specific reformatting.

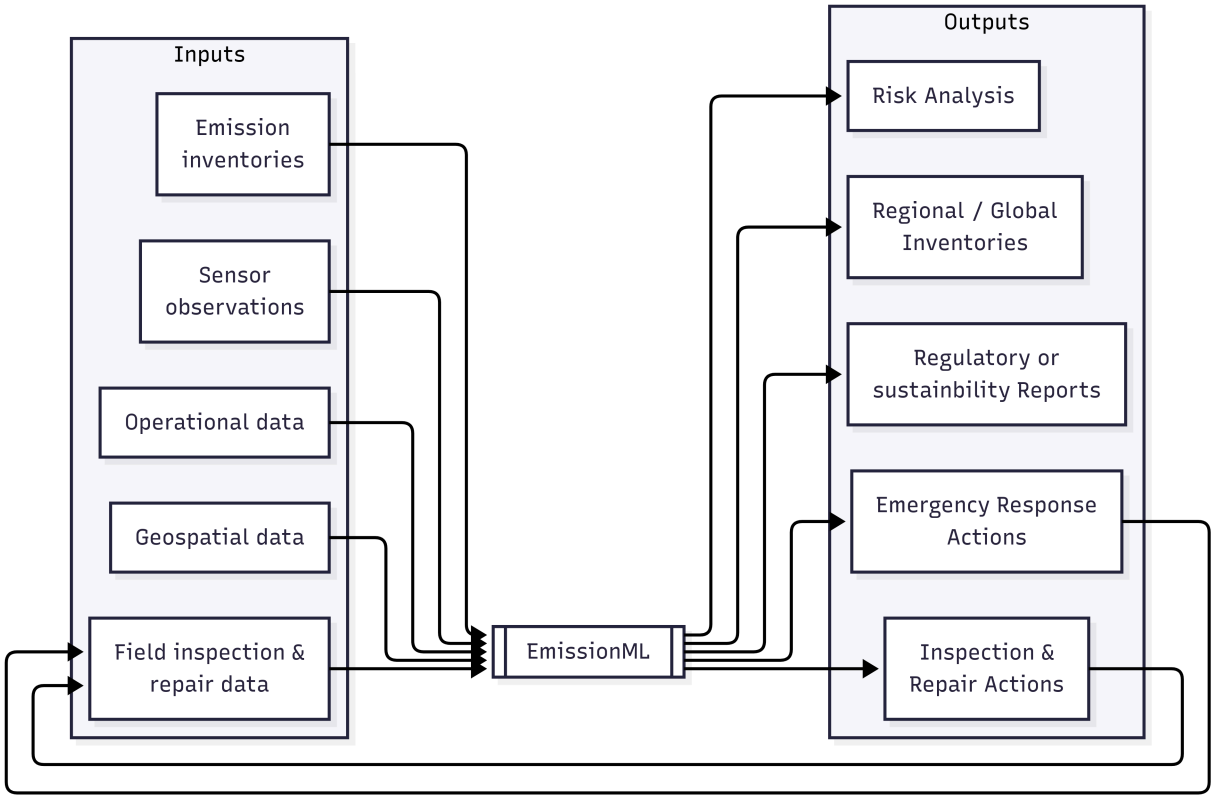



**Figure 1** — EmissionML as an intermediary data model: write once, use many times

EmissionML does not replace existing sensing technologies, estimation methods, or reporting tools. It provides the standards-grounded data foundation that enables them to interoperate.

## 8.4. Applicability Across Emission Domains

Although early development of EmissionML has been informed by oil and gas methane monitoring — where regulatory pressure and sensing technology are most advanced — the conceptual model is domain-neutral. The core abstraction of an emission event (a source releasing a substance through a mechanism, over a time interval, with a determined quantity) applies wherever substances are released into the atmosphere. Examples include:

- **Emergency management.** Wildfires, industrial accidents, and volcanic events produce large, episodic emissions whose source geometry, temporal bounds, and quantities must be characterized rapidly from heterogeneous observations (satellite imagery, ground sensors, dispersion models) and communicated across jurisdictions.
- **Landfill gas emissions.** Landfills emit methane and other gases through diffuse and point sources that vary with waste composition, cover integrity, and weather conditions. Standardized emission event records enable consistent monitoring and reporting across sites and regulatory regimes.

- **Permafrost thaw.** Thawing permafrost releases methane and carbon dioxide through mechanisms that are spatially diffuse and temporally variable. Reconciling field measurements, remote sensing, and biogeochemical models requires a common event model with explicit uncertainty provenance.
- **Agriculture.** Livestock, rice cultivation, and fertilizer application produce emissions through distinct mechanisms at varying spatial scales. A shared ontology enables aggregation from field-level observations to regional inventories.
- **Wastewater treatment.** Treatment processes emit methane, nitrous oxide, and other gases through identifiable mechanisms at known facilities, requiring the same source–event–observation linkage that EmissionML provides.
- **Public safety and hazardous releases.** Chemical spills, industrial gas leaks, and radiological dispersal events (e.g., dirty bombs) require rapid characterization of what substance was released, from where, in what quantity, and over what time interval — the same source–mechanism–quantity–temporal structure that EmissionML defines. A standardized event model enables interoperable data exchange across first responders, environmental agencies, and public health authorities.

The EmissionML conceptual model accommodates these domains without modification to the core classes or relationships. Domain-specific extensions — such as specialized Source Type or Mechanism vocabularies — can be defined as profiles of the core model.

## 8.5. Use Cases

Table 1 summarizes the key use cases that motivate the design of EmissionML.

**Table 1** — EmissionML use cases

| USE CASE | PROBLEM | EMISSIONML SOLUTION |
|---|---|---|
| Cross-vendor sensor integration | Proprietary payloads and inconsistent data models require bespoke integrations for each sensor system, increasing cost and preventing plug-and-play deployment. | Normalizes observation payloads to a consistent ontology, eliminating vendor-specific integration overhead. |
| Emission event attribution and validation | Satellite-detected concentration anomalies cannot be programmatically linked to specific sources (e.g., facilities, landfill cells, wildfire perimeters) when source feature definitions are inconsistent across systems. | Links Observations to Source Features and Emission Events with an auditable validation chain, reducing attribution uncertainty. |
| Regulatory and voluntary reporting | Operators reformat the same emission data for each reporting framework (EPA, EU Methane Regulation, OGMP 2.0, | Provides a single standards-conformant emission event record that can be |

| USE CASE | PROBLEM | EMISSIONML SOLUTION |
|---|---|---|
| | MiQ), duplicating effort and introducing transcription errors. | automatically transformed into multiple reporting formats. |
| Real-time operational response | Control rooms and emergency response centres receive fragmented event feeds from multiple systems, delaying root-cause diagnosis and coordinated action. | Delivers a normalized, brokerable event stream with consistent source and quantity semantics across platforms. |
| Carbon-market quantification and verification | Emission reduction calculations are opaque, with no standard mechanism to encode uncertainties or link quantities to their determination procedures. | Encodes emission quantities with uncertainties and provenance metadata, increasing credit integrity and reducing verification cost. |
| Emissions simulation interoperability | Simulation tools use incompatible models, limiting cross-study comparability and reuse of inputs and outputs. | Represents simulation inputs and outputs as standardized Emission Events, enabling benchmarking and reproducibility. |
| Financial risk and ESG analysis | ESG risk assessments depend on inconsistent, non-comparable emissions disclosures that cannot be reliably ingested into financial risk models. | Structures emissions data with standardized semantics and uncertainty metadata for direct ingestion into risk and ESG scoring models. |

## 8.6. Further Reading

The rationale, use cases, and design principles summarized in this clause are discussed in detail in the OGC EmissionML Discussion Paper OGC 25-023.

9

# EMISSION EVENT DATA MODEL

# 9 EMISSION EVENT DATA MODEL

## 9.1. Overview

The EmissionML conceptual model is described as a class diagram in Figure 2. The schema is fully described in x.x.x.

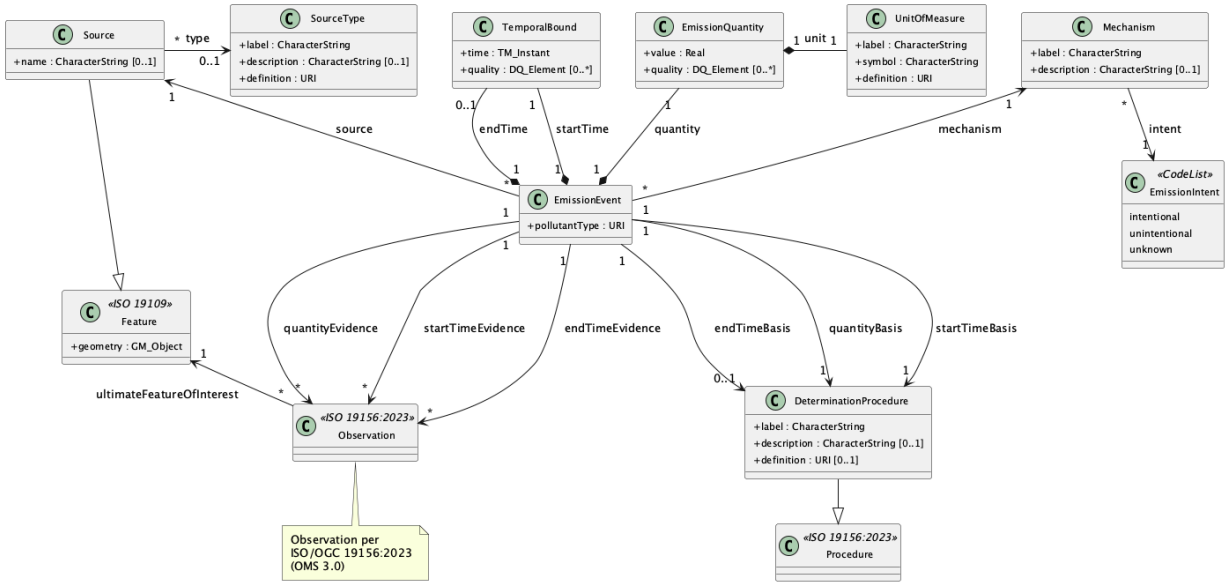
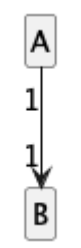



**Figure 2** — EmissionML Abstract Conceptual Model

**Table 2** — UML relationship notation used in this standard

| SYMBOL | NAME | MEANING |
|---|---|---|
| | Composition | The target is owned by and cannot exist without the source (e.g., EmissionEvent owns its EmissionQuantity and TemporalBound instances) |

| SYMBOL | NAME | MEANING |
|---|---|---|
|  | Association | A directional reference between independent classes (e.g., EmissionEvent refers to a Source) |

Use the following table for Requirements Classes.

**Table 3** — Requirements class — Emission Event Conceptual Data Model Core

| **Emission Event Conceptual Data Model Core** | |
|---|---|
| http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core | |
| Target type | Data Model |
| Dependency | http://www.opengis.net/doc/as/om/3.0/req/obs-basic |
| **Requirement 1** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/emission-event<br>This requirement defines the Emission Event. |
| **Requirement 2** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/source<br>This requirement defines the Source. |
| **Requirement 3** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/source-type<br>This requirement defines the Source Type. |
| **Requirement 4** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/mechanism<br>This requirement defines the Mechanism. |
| **Requirement 5** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/emission-quantity<br>This requirement defines the Emission Quantity. |
| **Requirement 6** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/unit-of-measure<br>This requirement defines the Unit of Measure. |
| **Requirement 7** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/temporal-bound<br>This requirement defines the Temporal Bound. |
| **Requirement 8** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/observation<br>This requirement defines the Observation. |
| **Requirement 9** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/emission-intent<br>This requirement defines the Emission Intent. |

| | |
|---|---|
| **Requirement 10** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/determination-procedure<br>This requirement defines the Determination Procedure. |
| **Requirement 11** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/feature<br>This requirement defines the Feature. |
| **Requirement 12** | http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/procedure<br>This requirement defines the Procedure. |

## 9.2. Emission Event Class

An Emission Event represents the release of a substance into the atmosphere over a non-zero duration. It is characterized by its start time and end time, the Source Feature from which the substance originated, and the identity and quantity of the emitted substance.

**NOTE 1:** Conceptually, an Emission Event corresponds to an occurrent in upper-ontology terms, as it unfolds over time and has temporal parts.

**NOTE 2:** For reporting or accounting purposes, a long-running emission event is often segmented into multiple time-bounded emission events (e.g., annual records). This segmentation reflects reporting conventions and does not imply physical cessation or re-initiation of the emission.

**Table 4** — Requirement 1 — Emission Event Class

| | |
|---|---|
| **Requirement 1: Emission Event Class** | /req/datamodel/core/emission-event<br><br>Each Emission Event class shall have the mandatory attributes listed in Table 5. Each Emission Event class shall have the direct relation between an Emission Event class and other classes listed in Table 6. |

**Table 5** — Attributes of an Emission Event class

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| pollutant Type | The URI linking to an external definition of the pollutantType of the Emission Event. | URI | 1 (mandatory) |

**Table 6** — Direct relation between an Emission Event class and other class types

| ROLE | DESCRIPTION | TARGET CLASS TYPE | MULTIPLICITY AND USE |
|---|---|---|---|
| hasMechanism | The mechanism by which the emission occurred. | Mechanism | 1 (mandatory) |
| hasSource | The source from which the emission originated. | Source | 1 (mandatory) |
| startTime | The start time of the emission event. | TemporalBound | 1 (mandatory) |
| endTime | The end time of the emission event. | TemporalBound | 0..1 (optional) |
| hasQuantity | The quantity of the substance emitted. | EmissionQuantity | 1 (mandatory) |
| hasStartTimeBasis | The procedure used to determine the start time. | Determination Procedure | 1 (mandatory) |
| hasEndTimeBasis | The procedure used to determine the end time. | Determination Procedure | 0..1 (optional) |
| hasQuantityBasis | The procedure used to determine the emission quantity. | Determination Procedure | 1 (mandatory) |
| hasStartTime Evidence | Observations providing evidence for the start time determination. | Observation | 0..* (optional) |
| hasEndTimeEvidence | Observations providing evidence for the end time determination. | Observation | 0..* (optional) |
| hasQuantityEvidence | Observations providing evidence for the quantity determination. | Observation | 0..* (optional) |

## 9.3. Feature Class

A Feature is an abstraction of real world phenomena (Source: ISO 19101-1:2014). In EmissionML, Feature serves as the base class for entities that can be the ultimate feature of interest of an Observation. A Feature's `geometry` attribute provides the geospatial reference for the phenomenon it represents.

**Table 7** — Requirement 11 — Feature Class

| | |
|---|---|
| Requirement 11: Feature Class | /req/datamodel/core/feature<br>+ Each Feature class shall have the mandatory attributes listed in Table 8. |

**Table 8** — Attributes of a Feature class

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| geometry | The spatial geometry of the feature. | GM_Object | 1 (mandatory) |

**Table 9** — Direct relation between a Feature class and other class types

| ROLE | DESCRIPTION | TARGET CLASS TYPE | MULTIPLICITY AND USE |
|---|---|---|---|
| - | No direct relations are defined for Feature in the core model (it is referenced by Source via inheritance and by Observation via ultimateFeatureOfInterest). | - | - |

## 9.4. Source Class

A Source represents the feature from which an emission originates. A Source is an ISO 191xx feature whose geometry provides the geospatial reference for one or more Emission Events. A Source is characterised by its own identity, geometry, intended function, and lifecycle, and is explicitly agnostic to the Mechanism by which any associated Emission Event occurs.

**NOTE:** A Source is a continuant that may exist independently of any emission, whereas an Emission Event is an occurrent. The identity and lifecycle of a Source are independent of the existence, absence, or frequency of Emission Events.

**Table 10** — Requirement 2 — Source Class

| | |
|---|---|
| Requirement 2: Source Class | /req/datamodel/core/source<br>+ Each Source class shall inherit from Feature. Each Source class shall have the optional attributes listed in Table 11. Each Source class shall have the direct relation between a Source class and other classes listed in Table 12. |

**Table 11** — Attributes of a Source class (own attributes; geometry is inherited from Feature)

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| name | A human-readable name for the source. | CharacterString | 0..1 (optional) |

**Table 12** — Direct relation between a Source class and other class types

| ROLE | DESCRIPTION | TARGET CLASS TYPE | MULTIPLICITY AND USE |
|---|---|---|---|
| hasType | The type classification of the source. | SourceType | 0..1 (optional) |

## 9.5. Source Type Class

A Source Type classifies a Source by its category.

**Table 13** — Requirement 3 — Source Type Class

| | |
|---|---|
| Requirement 3: Source Type Class | /req/datamodel/core/source-type<br>+ Each Source Type class shall have the mandatory attributes listed in Table 14. |

**Table 14** — Attributes of a Source Type class

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| label | A human-readable label for the source type. | CharacterString | 1 (mandatory) |
| description | A detailed description of the source type. | CharacterString | 0..1 (optional) |
| definition | The URI linking to an external definition of the source type. | URI | 1 (mandatory) |

**Table 15** — Direct relation between a Source Type class and other class types

| ROLE | DESCRIPTION | TARGET CLASS TYPE | MULTIPLICITY AND USE |
|---|---|---|---|
| - | No direct relations are defined for SourceType in the core model (it is referenced by Source). | - | - |

## 9.6. Mechanism Class

A Mechanism is a classification of the specific physical, chemical, or operational process that constitutes the direct cause or pathway for an emission. It is defined independently of the

Source—including its identity, structure, or purpose—allowing Mechanisms to be consistently reused across different types of Source.

**Table 16** — Requirement 4 — Mechanism Class

| | |
|---|---|
| Requirement 4: Mechanism Class | /req/datamodel/core/mechanism<br>+ Each Mechanism class shall have the mandatory attributes listed in Table 17. Each Mechanism class shall have the direct relation listed in Table 18. |

**Table 17** — Attributes of a Mechanism class

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| label | A human-readable label for the mechanism. | CharacterString | 1 (mandatory) |
| description | A detailed description of the mechanism. | CharacterString | 0..1 (optional) |

**Table 18** — Direct relation between a Mechanism class and other class types

| ROLE | DESCRIPTION | TARGET CLASS TYPE | MULTIPLICITY AND USE |
|---|---|---|---|
| has Intent | The emission intent associated with the mechanism. | EmissionIntent | 1 (mandatory) |

## 9.7. Emission Quantity Class

An Emission Quantity is the physical quantity representing the total mass of substance released to the atmosphere during a defined Emission Event or over a specified time interval.

**Table 19** — Requirement 5 — Emission Quantity Class

| | |
|---|---|
| Requirement 5: Emission Quantity Class | /req/datamodel/core/emission-quantity<br>+ Each Emission Quantity class shall have the mandatory attributes listed in Table 20. Each Emission Quantity class shall have the direct relation between an Emission Quantity class and other classes listed in Table 21. |

**Table 20** — Attributes of an Emission Quantity class

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| value | The numeric value of the emission quantity. | Real | 1 (mandatory) |
| quality | A data quality element describing the quality of the emission quantity value. | DQ_Element | 0..* (optional) |

**Table 21** — Direct relation between an Emission Quantity class and other class types

| ROLE | DESCRIPTION | TARGET CLASS TYPE | MULTIPLICITY AND USE |
|---|---|---|---|
| has Unit | The unit of measure associated with the emission quantity. | UnitOfMeasure | 1 (mandatory) |

# 9.8. Unit of Measure Class

A Unit of Measure is a reference quantity chosen from a unit equivalence group (Source: ISO/OGC 19156:2023)

**Table 22** — Requirement 6 — Unit of Measure Class

| | |
|---|---|
| Requirement 6: Unit of Measure Class | /req/datamodel/core/unit-of-measure<br>+ Each Unit of Measure class shall have the mandatory attributes listed in Table 23. |

**Table 23** — Attributes of a Unit of Measure class

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| label | A human-readable label for the unit of measure. | CharacterString | 1 (mandatory) |
| symbol | A short symbol representing the unit of measure. | CharacterString | 1 (mandatory) |
| definition | The URI linking to an external definition of the unit of measure. | URI | 1 (mandatory) |

**Table 24** — Direct relation between a Unit of Measure class and other class types

| ROLE | DESCRIPTION | TARGET CLASS TYPE | MULTIPLICITY AND USE |
|---|---|---|---|
| - | No direct relations are defined for UnitOfMeasure in the core model. | - | - |

## 9.9. Temporal Bound Class

A Temporal Bound is a value object representing a temporal determination (e.g., start time or end time) that MAY have associated data quality metadata. The determination procedure and supporting evidence for a Temporal Bound are tracked at the Emission Event level.

**Table 25** — Requirement 7 — Temporal Bound Class

| | |
|---|---|
| Requirement 7: Temporal Bound Class | /req/datamodel/core/temporal-bound<br>+ Each Temporal Bound class shall have the mandatory attributes listed in Table 26. |

**Table 26** — Attributes of a Temporal Bound class

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| time | The time instant value of this Temporal Bound. | TM_Instant | 1 (mandatory) |
| quality | A data quality element describing the quality of the temporal bound. | DQ_Element | 0..* (optional) |

**Table 27** — Direct relation between a Temporal Bound class and other class types

| ROLE | DESCRIPTION | TARGET CLASS TYPE | MULTIPLICITY AND USE |
|---|---|---|---|
| - | No direct relations are defined for TemporalBound in the core model (determination procedure and evidence are tracked at the EmissionEvent level). | - | - |

## 9.10. Observation Class

An Observation is an act carried out by an observer to determine the value of an observable property of an object (feature-of-interest) by using a procedure, with the value provided as the

result (Source: ISO/OGC 19156:2023). In EmissionML, an Observation is evidence that may support an Emission Quantity or Temporal Bound determination.

**NOTE:** The Ultimate Feature of Interest of an Observation identifies a Feature — which may be a Source or another feature at a different spatial scale — associated with the Emission Event. It is common for a single Emission Event to be observed by multiple Observations whose Ultimate Features of Interest are defined at different spatial scales. For example, a satellite remote sensing image Observation typically has an area-based Ultimate Feature of Interest (e.g., a facility area or region), whereas the same Emission Event may also be observed by a person carrying an Optical Gas Imaging camera, for which the Ultimate Feature of Interest is a specific Source such as a component of an item of equipment from which pollutants are released.

**Table 28** — Requirement 8 — Observation Class

| | |
|---|---|
| Requirement 8: Observation Class | /req/datamodel/core/observation<br>+ Each Observation class shall be an ISO/OGC 19156:2023 (OGC OMS 3.0) Observation. |

## 9.11. Emission Intent Enumeration

Emission Intent is a code list describing whether an emission is intentional, unintentional, or unknown.

**Table 29** — Requirement 9 — Emission Intent Enumeration

| | |
|---|---|
| Requirement 9: Emission Intent Enumeration | /req/datamodel/core/emission-intent<br>+ The Emission Intent enumeration shall support the enumeration values listed in Table 30. |

**Table 30** — Enumeration values of Emission Intent

| NAME | DEFINITION |
|---|---|
| intentional | The emission occurred intentionally. |
| unintentional | The emission occurred unintentionally. |
| unknown | The intent of the emission is unknown. |

## 9.12. Procedure Class

A Procedure is a description of steps performed (Source: ISO/OGC 19156:2023). In EmissionML, Procedure serves as the base class for process descriptions such as DeterminationProcedure.

**Table 31** — Requirement 12 — Procedure Class

| | |
|---|---|
| Requirement 12: Procedure Class | /req/datamodel/core/procedure<br>+ Each Procedure class shall be an ISO/OGC 19156:2023 (OGC OMS 3.0) Procedure. |

**Table 32** — Attributes of a Procedure class

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| - | No own attributes are defined for Procedure in the core model (see the standard for complete specification). | - | - |

**Table 33** — Direct relation between a Procedure class and other class types

| ROLE | DESCRIPTION | TARGET CLASS TYPE | MULTIPLICITY AND USE |
|---|---|---|---|
| - | No direct relations are defined for Procedure in the core model (it is referenced by DeterminationProcedure via inheritance). | - | - |

## 9.13. Determination Procedure Class

A Determination Procedure describes the steps performed to determine the quantity or temporal bounds of an Emission Event. DeterminationProcedure inherits from Procedure, aligning it with the OMS 3.0 process model.

**Table 34** — Requirement 10 — Determination Procedure Class

| | |
|---|---|
| Requirement 10: Determination Procedure Class | /req/datamodel/core/determination-procedure<br>+ Each Determination Procedure class shall inherit from Procedure. Each Determination Procedure class shall have the mandatory attributes listed in Table 35. |

**Table 35** — Attributes of a Determination Procedure class (own attributes)

| NAMES | DEFINITION | DATA TYPES AND VALUES | MULTIPLICITY AND USE |
|---|---|---|---|
| label | A human-readable label for the determination procedure. | CharacterString | 1 (mandatory) |
| description | A detailed description of the determination procedure. | CharacterString | 0..1 (optional) |
| definition | The URI linking to an external definition of the determination procedure. | URI | 0..1 (optional) |

A

# ANNEX A (INFORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)

# A ANNEX A (INFORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)

## A.1. Abstract tests for Emission Event Conceptual Data Model Core

### A.1.1. Emission Event Conceptual Data Model Core

**Table A.1** — Conformance test — Emission Event Conceptual Data Model Core

| | |
|---|---|
| Conformance class | /conf/datamodel/core |
| Requirements | /req/datamodel/core |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

### A.1.2. Emission Event

**Table A.2** — Conformance test — Emission Event

| | |
|---|---|
| Conformance class | /conf/datamodel/core/emission-event |
| Requirements | /req/datamodel/core/emission-event |

| | |
|---|---|
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

### A.1.3. Source

**Table A.3** — Conformance test — Source

| | |
|---|---|
| Conformance class | /conf/datamodel/core/source |
| Requirements | /req/datamodel/core/source |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

### A.1.4. Source Type

**Table A.4** — Conformance test — Source Type

| | |
|---|---|
| Conformance class | /conf/datamodel/core/source-type |
| Requirements | /req/datamodel/core/source-type |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

### A.1.5. Mechanism

**Table A.5** — Conformance test — Mechanism

| | |
|---|---|
| Conformance class | /conf/datamodel/core/mechanism |
| Requirements | /req/datamodel/core/mechanism |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

## A.1.6. Emission Quantity

**Table A.6** — Conformance test — Emission Quantity

| | |
|---|---|
| Conformance class | /conf/datamodel/core/emission-quantity |
| Requirements | /req/datamodel/core/emission-quantity |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

## A.1.7. Unit of Measure

**Table A.7** — Conformance test — Unit of Measure

| | |
|---|---|
| Conformance class | /conf/datamodel/core/unit-of-measure |
| Requirements | /req/datamodel/core/unit-of-measure |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

## A.1.8. Temporal Bound

**Table A.8** — Conformance test — Temporal Bound

| | |
|---|---|
| Conformance class | /conf/datamodel/core/temporal-bound |
| Requirements | /req/datamodel/core/temporal-bound |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

## A.1.9. Observation

**Table A.9** — Conformance test — Observation

| | |
|---|---|
| Conformance class | /conf/datamodel/core/observation |
| Requirements | /req/datamodel/core/observation |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

## A.1.10. Emission Intent

**Table A.10** — Conformance test — Emission Intent

| | |
|---|---|
| Conformance class | /conf/datamodel/core/emission-intent |
| Requirements | /req/datamodel/core/emission-intent |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |

| | |
|---|---|
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

## A.1.11. Determination Procedure

**Table A.11** — Conformance test — Determination Procedure

| | |
|---|---|
| Conformance class | /conf/datamodel/core/determination-procedure |
| Requirements | /req/datamodel/core/determination-procedure |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

## A.1.12. Feature

**Table A.12** — Conformance test — Feature

| | |
|---|---|
| Conformance class | /conf/datamodel/core/feature |
| Requirements | /req/datamodel/core/feature |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

## A.1.13. Procedure

**Table A.13** — Conformance test — Procedure

| | |
|---|---|
| Conformance class | /conf/datamodel/core/procedure |

| | |
|---|---|
| Requirements | /req/datamodel/core/procedure |
| Test purpose | Verify that all requirements from the requirements class have been fulfilled. |
| Test method | Inspect the documentation of the application, schema or profile. |
| Test type | Capability |

B

# ANNEX B (INFORMATIVE) RELATIONSHIP WITH OGC FEATURES AND OBSERVATIONS

# B ANNEX B (INFORMATIVE) RELATIONSHIP WITH OGC FEATURES AND OBSERVATIONS

## B.1. Overview

EmissionML is a domain-specific conceptual model that builds on two foundational OGC concepts: **Features** (ISO 19109 General Feature Model) and **Observations** (ISO/OGC 19156:2023, OMS 3.0). This annex explains how EmissionML relates to each and why explicit domain modeling is preferred over generic usage.

Figure B.1 illustrates the conceptual layering of EmissionML on top of these foundational standards.

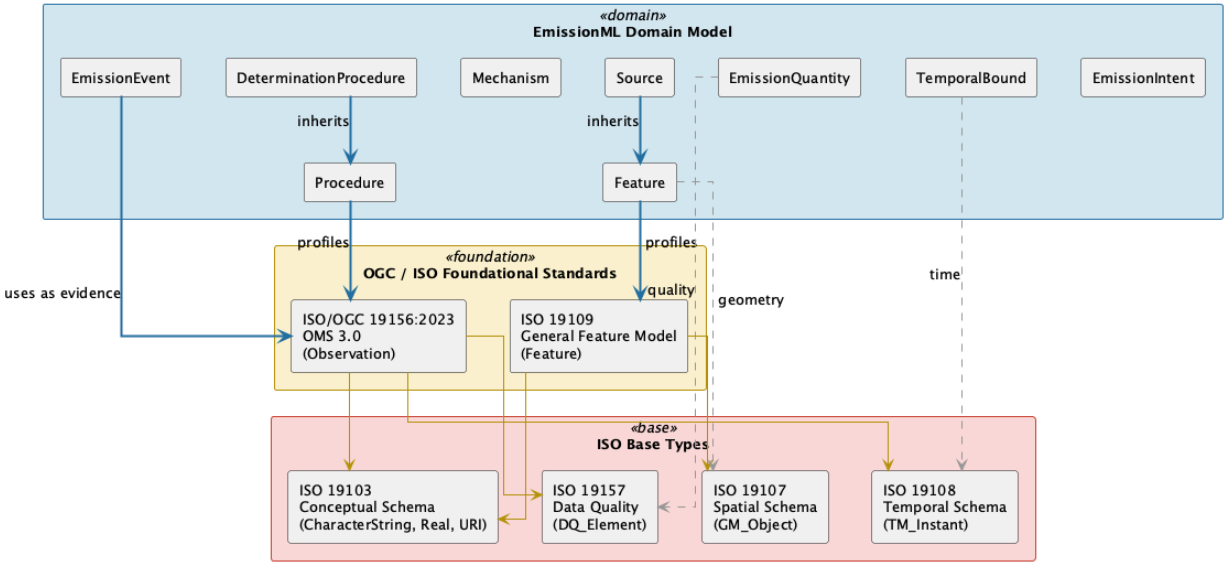



**Figure B.1** — Conceptual layering of EmissionML on foundational ISO/OGC standards

## B.2. EmissionML and OGC Features

The OGC General Feature Model (ISO 19109) defines a *Feature* as an abstraction of a real-world phenomenon — any entity with an identity, properties, and optionally a geometry. EmissionML makes this alignment explicit by introducing a concrete **Feature** class (stereotyped `<<ISO 19109>>`) into the conceptual model.

The Feature class carries a mandatory `geometry` attribute of type `GM_Object` and serves as the base class for domain-specific feature types. **Source** inherits from Feature, gaining its geometry by inheritance. This structural inheritance — rather than a notional stereotype — makes the ISO 19109 alignment unambiguous and machine-verifiable.

Because Feature is concrete, an Observation's `ultimateFeatureOfInterest` can reference any Feature — not only a Source. This enables observations whose feature of interest is defined at a different spatial scale (e.g., a facility area, a region, or a monitoring zone) to participate in the evidence chain without being forced into the Source class.

**EmissionEvent** is also a real-world phenomenon — an occurrent that unfolds over time. In principle, an Emission Event *could* be modeled as a generic Feature. However, EmissionML gives it explicit structure: temporal bounds (start time, end time), a quantity of emitted substance, a mechanism describing the physical or operational process, an intent classification, and an evidence chain linking the event to supporting Observations.

Without this explicit modeling, a Source would be indistinguishable from any other Feature, and an emission event would be "just another Feature" with ad-hoc properties. The semantic relationships that give emission data its meaning — `hasSource`, `hasMechanism`, `hasQuantity`, `hasStartTimeEvidence` — would be lost or require convention-based interpretation.

## B.3. EmissionML and OMS 3.0 Observations

OMS 3.0 (ISO/OGC 19156:2023) defines an Observation as an act of determining the value of an observable property. An Observation has a well-defined structure: a feature of interest, an observed property, a procedure, and a result.

EmissionML incorporates OMS 3.0 Observation **by reference** (Requirement 8) — it does not redefine or extend the Observation class. In EmissionML, Observations serve as **evidence** supporting emission quantity and temporal bound determinations, linked through the `hasQuantityEvidence`, `hasStartTimeEvidence`, and `hasEndTimeEvidence` associations.

One might ask: *Why not model the entire emission event as an Observation result?* Because an Emission Event is richer than what a single Observation captures. An Emission Event has a source, a mechanism, an intent, multiple temporal bounds, and can be supported by *multiple* Observations at different spatial and temporal scales. For example:

- A satellite remote sensing Observation may detect a methane plume over a broad area.

- An Optical Gas Imaging (OGI) camera Observation may pinpoint the leak to a specific component.
- A continuous emissions monitoring system (CEMS) Observation may quantify the emission rate over time.

All three Observations can serve as evidence for a single Emission Event, each contributing to the determination of quantity, start time, or end time. The Observation's `ultimateFeatureOfInterest` references a Feature — which may be a Source or any other Feature at a different spatial scale. The satellite Observation's feature of interest may be a Feature representing an entire facility, while the OGI Observation's feature of interest is a Source representing a single valve.

## B.4. EmissionML and OMS 3.0 Procedures

OMS 3.0 defines a **Procedure** as "a description of steps performed" and introduces **ObservingProcedure** as a specialization that describes steps performed to determine the value of an observable property. EmissionML follows this same specialization pattern.

The EmissionML conceptual model introduces a **Procedure** class (stereotyped `<<ISO 19156:2023>>`) as the base class for process descriptions. **DeterminationProcedure** inherits from Procedure and describes the steps performed to determine a value — such as an emission quantity or temporal bound. This structural inheritance mirrors how ObservingProcedure specializes Procedure in OMS 3.0.

By making Procedure an explicit base class rather than a notional reference, EmissionML ensures that its process descriptions are structurally aligned with OMS 3.0 — not merely by naming convention, but by class hierarchy. This parallels the Feature/Source pattern: just as Source inherits from Feature to make ISO 19109 alignment concrete, DeterminationProcedure inherits from Procedure to make OMS 3.0 alignment concrete.

## B.5. Why Explicit Domain Modeling

**The alternative.** One could model everything with generic Features and Observations — Sources as plain Features, emission data as Observation results, and relationships as ad-hoc properties. This approach uses only existing OGC standards without defining new classes.

**The problem.** Generic modeling loses domain semantics. Without EmissionML's explicit classes, systems cannot:

- distinguish a Source from any other Feature,
- enforce that an emission has a Mechanism and an Intent,

- trace the evidence chain from Observation to Emission Quantity to Emission Event,
- validate that required relationships are present (e.g., every Emission Event must have a Source, a Mechanism, and a Determination Procedure).

**What explicit modeling provides:**

| | |
|---|---|
| Interoperability | Systems exchanging EmissionML data share a common vocabulary. EmissionEvent, Source, Mechanism, and EmissionQuantity are unambiguous — they mean the same thing in every implementation. |
| Validation | Requirements can be formally specified and enforced. For example, Requirement 1 mandates that every Emission Event has a Source, a Mechanism, start time, quantity, and their corresponding Determination Procedures. |
| Usability | Domain experts work with emission concepts directly — *emission events*, *sources*, *mechanisms* — rather than mapping to and from generic abstractions. This reduces the cognitive burden and the risk of misinterpretation. |

**Analogy.** Just as CityGML defines explicit Building, Road, and Vegetation classes rather than using generic Features, EmissionML defines explicit emission domain classes. The General Feature Model provides the abstract foundation; EmissionML provides the domain-specific vocabulary that makes emission data interoperable and actionable.

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